# BCSE204L Design and Analysis of Algorithms

Module-2

Topic-5:Backtracking

# Backtracking Basics

- Another algorithm design paradigm
- Uses brute-force
- Try all solutions and pick the desired ones
- DP also does the same??
- But DP is for optimization problems
- Backtracking is NOT for optimization problems

# Backtracking

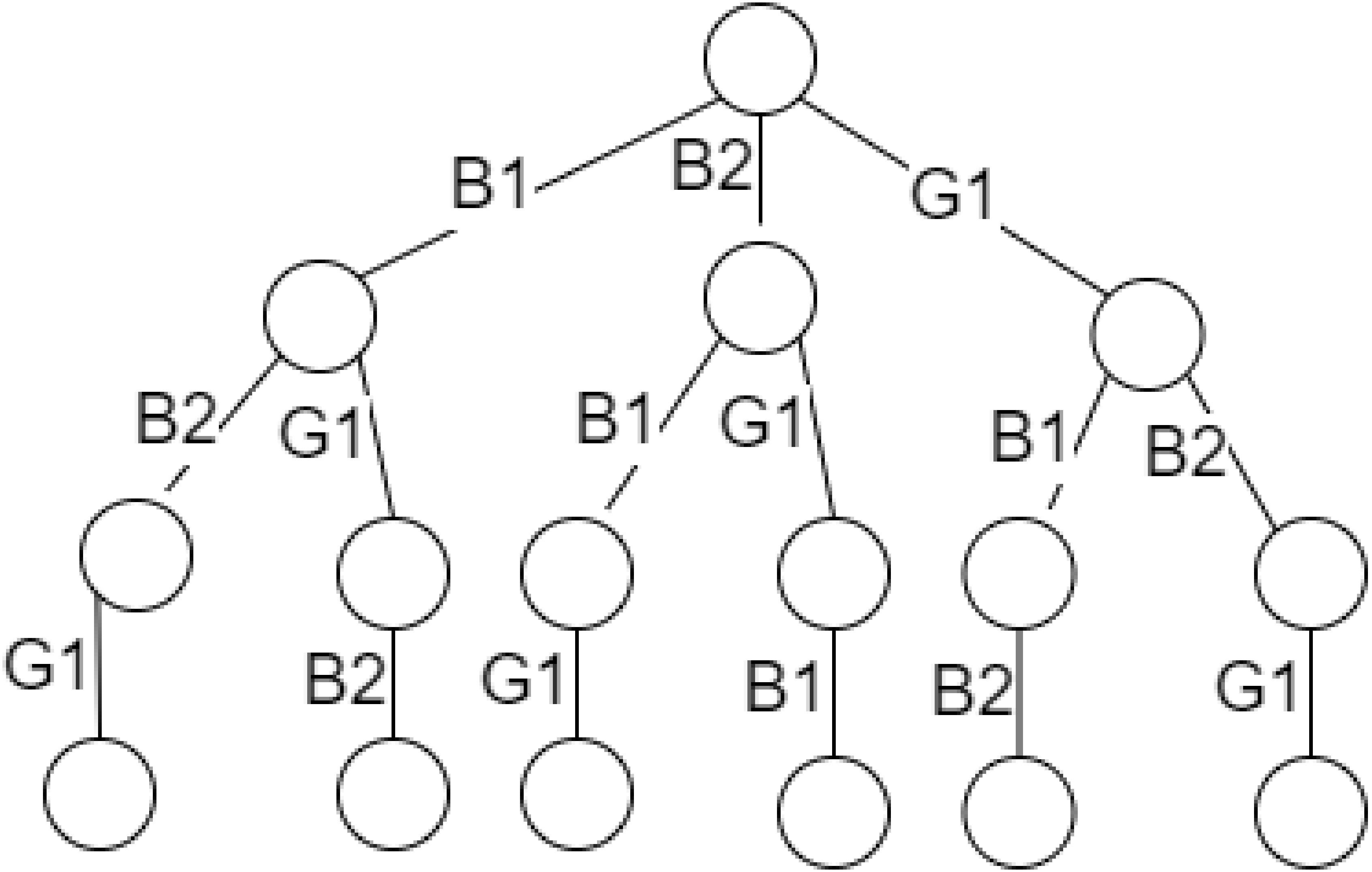
- Is used when a problem has multiple solutions and we want all of them
- We use a state space tree to represent the solutions
- Many a times the problem would be to find only those solutions which satisfy some constraints
- In that case the we kill the nodes by applying the bounding function and then backtrack

# An Example: No constraints

- We have 3 children:
- 1 Girl (G1) and 2 Boys (B1 and B2)
- We have 3 chairs
- In how many ways the children can sit in the chairs
- There are $3! = 6$ possible solutions
- We want all, can be represented as a state space tree

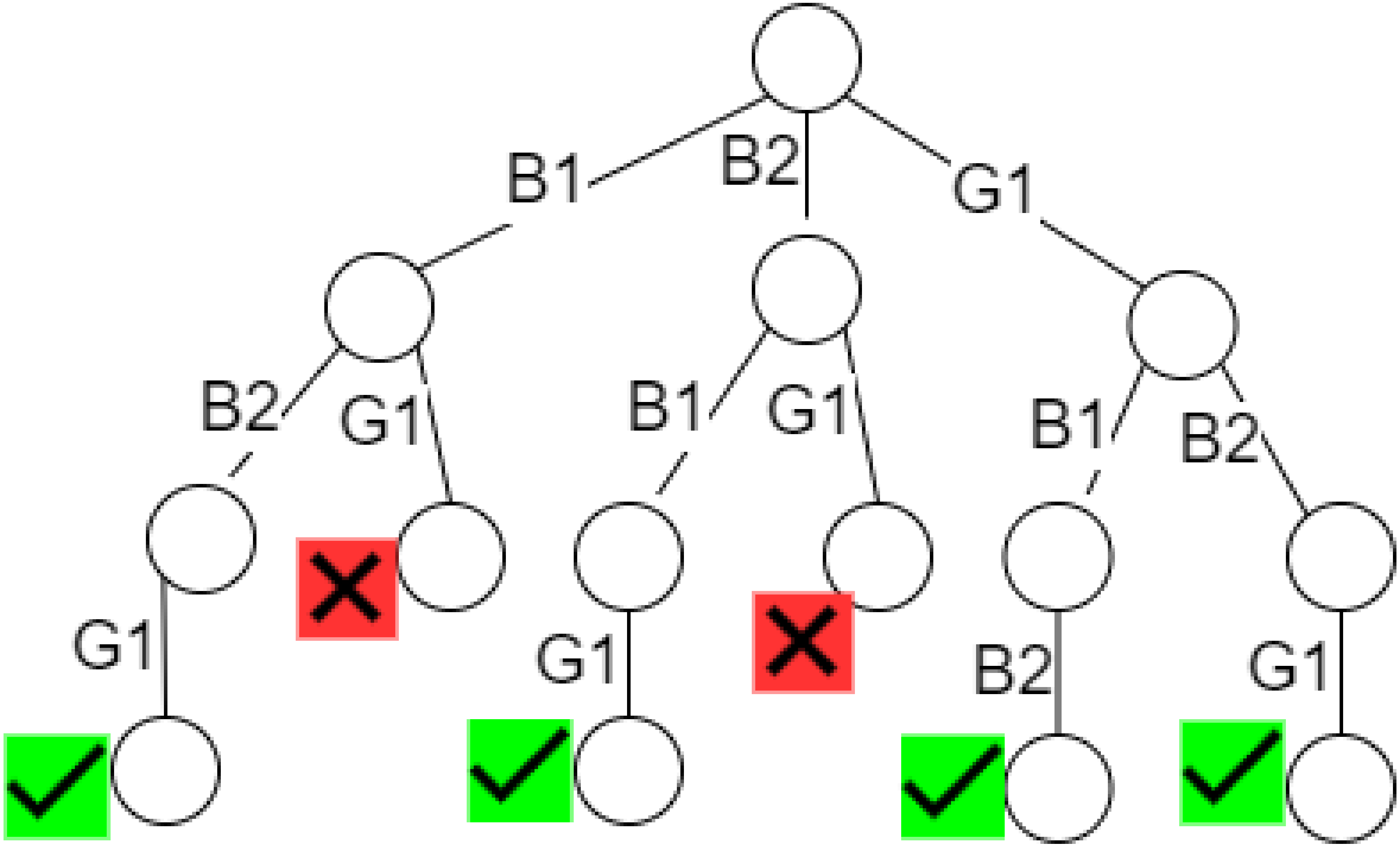
| Chair 1 | Chair 2 | Chair 3 |
|---|---|---|

G1 B1 B2



# Example: with constraints

- Impose the constraint that the girl cannot sit in the middle
- We develop the state space tree as in the previous case, but we will apply the bounding function and kill the nodes which do not proceed to a solution
- Total 4 solutions, we want all


B1
B2
G1
B2
G1
B1
G1
B1
B2
G1
G1
B2
G1

Thank you..!!